

AST Prep

A bilingual practice companion for recreational avalanche-safety training (AST 1 & AST 2). This manual explains how to run classes, manage learners, control what each class sees, and read progress.

App: avalanche-safety-training-prep.vercel.app

Languages: English / Français

Audience: Administrators & instructors

This guide is for the people who run AST Prep for a group — instructors, education leads, and administrators. Learners don't need it; they just open the app, join their class, and practise. Everything below is reached from the in-app admin console.

Safety notice. AST Prep is a study-and-practice aid. It does **not** replace professional avalanche training (AST 1 / AST 2), the current regional avalanche bulletin, or real-time judgement in the field. It is an independent tool and is not affiliated with or endorsed by Avalanche Canada, Parks Canada, or the Canadian Avalanche Association; it references their public frameworks (danger scale, avalanche problems, ATES, Avaluator, OGRS) for educational purposes only.

1 Roles & access

Role	What they can do
Super-admin	The owner account. Full admin rights that can never be removed. Manages other admins.
Admin / instructor	Opens the admin console: create and manage classes, add/remove members, set access windows, control section visibility, view everyone's performance and leaderboards, and manage the question bank.
Member (learner)	A signed-in student in one or more classes. Practises tools, tracks their own progress, appears on their class leaderboard.
Guest	Someone using the app without signing in. Progress is saved only on their device. You can turn guest access off (see §7).

2 Signing in

Existing users

Sign in with a **magic link** (enter email, tap the link that arrives) or with **email + password**.

New learners

New accounts require **email + password + a class join code**. Without a valid, active code, sign-up is refused — this is how classes stay private.

Admins reach the console from the **Home** screen: when signed in as an admin, a **Site Admin** card appears under "Progress & tools." It has three areas — **Users**, **Question bank**, and **Sessions & access** (your class controls).

3 The learner's Home screen

Home is organized into three tappable **sections**. Tapping a section reveals its tools; each tool has a back button to its section.

Section	Contains
AST 1 Fundamentals	Slope-Angle, Danger-Rating, Bulletin, Terrain-Trap, ATES, Beacon, and the AST 1 practice exam.
AST 2 Advanced	Everything in AST 1 <i>plus</i> Snowpack-Test, Crystal-Card grain ID, Scenario Decisions, and the AST 2 practice exam.
Practical Planning	Trip Planner (Avaluator) and Trip & Field Checklists — useful at any level.

You decide which of these three sections each class can see (§6).

4 Managing classes

In the admin console, open **Sessions & access**. A "class" (called a *session* in the app) is a cohort of learners who share a join code.

Create a class

- 1 Under **New class**, type a human-readable label (e.g. "Feb 2026 AST 1 — Tuesday"). Tap **Create**.
- 2 The app generates a hard-to-guess 10-character **join code**. Share this code with that cohort only.
- 3 Learners create their account with their email, a password, and this code.

Add or remove members

Open a class and tap **Manage**. Add an existing user by email, or remove a member. Members can belong to more than one class.

Access windows

Control how long a learner keeps access:

- **Default** — a set number of weeks from sign-up (the global default is on the Sessions page; changing it flows to everyone on the default setting).
- **Unlimited** — never expires.
- **Until a date** — expires on a date you pick.

When access lapses, a learner can still see their history and delete their data, but can't record new practice until you extend them. Expired learners see a "contact your instructor" screen.

Activate / deactivate & the Legacy group

Deactivating a class blocks new sign-ups with its code (existing members keep access). A built-in **Legacy** group grandfathers any pre-existing users with unlimited access.

5 Tracking progress

Performance dashboard

Per-tool breakdowns plus, for the exams, a **Readiness** score, a **Mastery grid** (topic × difficulty), and **Calibration** (how well confidence matches correctness). Admins can view learners' analytics.

Leaderboards

Learners see their class leaderboard with a global toggle; admins can view any class or the global board. Appearing on a leaderboard is opt-in for learners.

6 Controlling what a class sees

You can show a class **any, some, or none** of the three Home sections — for example, an AST 1 cohort that should only see AST 1 material.

- **Per class**: open a class → **Manage** → **Sections visible to this class**. Toggle AST 1 / AST 2 / Practical (green ✓ = shown, grey × = hidden). Members see only what's on.
- **Default for new classes**: on the Sessions page, **Default sections for new classes** sets what freshly created classes start with. Changing it does not alter existing classes.

How it resolves: admins always see all three. A learner in more than one class sees the **union** of their classes' sections (a broader class grants more). Guests and anyone with no class see all three. If a learner's classes leave nothing visible, Home shows a "contact your instructor" note. Existing classes default to showing all three.

7 Guest access

By default anyone can practise as a guest (progress saved on their device only). To require every user to sign in, turn on **Require sign-in** at the top of the Sessions page. Turn it off to re-open guest access.

8 Tools reference

What each tool trains, so you can point learners at the right practice.

Tool	Section	Trains
Slope-Angle Trainer	AST 1	Estimating slope steepness and the avalanche-prone range.
Danger-Rating Trainer	AST 1	Reading the danger by elevation band and calling the rating.

Bulletin Trainer	AST 1	Interpreting a bulletin — rating meaning, problems, and management.
Terrain-Trap Trainer	AST 1	Recognizing terrain traps and consequences.
ATES Classifier	AST 1	Classifying terrain Simple / Challenging / Complex.
Beacon Search	AST 1	Companion-rescue transceiver search practice.
Crystal-Card Trainer	AST 2	Snow grain identification and sizing.
Snowpack-Test Interpreter	AST 2	Reading ECT / CT stability-test results.
Scenario Decisions	AST 2	Multi-step decision-making in realistic scenarios.
AST 1 / AST 2 Practice	1 2	Exam-style question banks with study, review, and adaptive modes.
Trip Planner	Practical	Avaluator: danger × terrain → trip recommendation.
Trip & Field Checklists	Practical	Editable planning / field checklists with printable trip sheet (PDF).

9 Question bank & language

Question bank overrides

The **Question bank** page lets admins adjust a question's correct answer or explanation without redeploying — useful for corrections or local emphasis.

Language

Every learner can switch between **English** and **French** from the top bar; their choice is remembered. All tools and content are fully bilingual.

10 Privacy & data

- Signed-in progress syncs to the learner's account; guest progress stays on the device.
- Every learner can **delete their data** or **delete their account** from the Account screen. Data is never removed unless the learner or an admin requests it.
- Access rules are enforced on the server; hiding a section is a visibility choice, not a data-security boundary.

11 Instructor tips

- Name classes so you can tell cohorts apart at a glance (term, level, day).
- For an AST 1-only cohort, hide AST 2 so learners aren't shown material beyond their course.
- Set the access window to match your program (e.g. the course plus a review period), then extend keen learners to unlimited.
- Use the Readiness score and Mastery grid to spot topics a group is weak on before an exam.
- Encourage the Trip Planner and Checklists as habits learners carry into the field — not just study tools.

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